Exploratory Data Analysis & Data Cleaning Report

Internship Track: Data Science Internship – Week 1, Task 3 | **Author:** Sayantani Adhikary

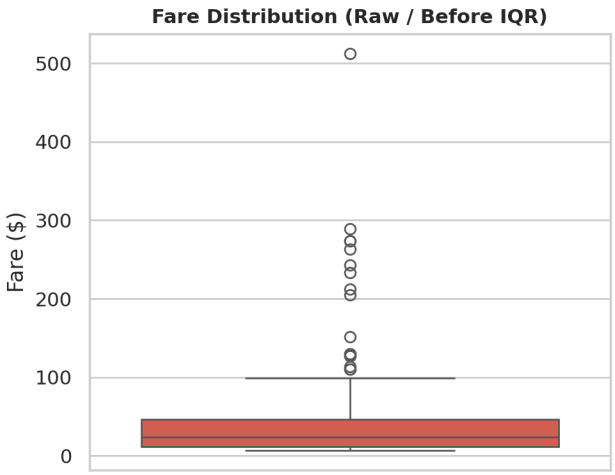
1. Dataset Overview

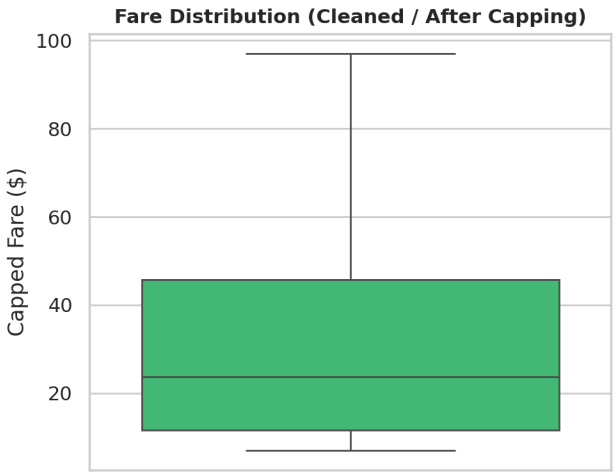
The dataset analyzed is the classic Titanic Passenger dataset containing demographic, socio-economic, and survival records across 150 passengers. The dataset features 9 primary variables including passenger class, age, gender, fare, family accompaniment, and embarkation port.

Feature	Type	Description	Initial Quality Status
PassengerId	Integer	Unique passenger identifier	100% complete, 2 duplicates
Survived	Integer	Target label (0 = No, 1 = Yes)	Binary classification label
Pclass	Integer	Ticket class (1st, 2nd, 3rd)	Ordinal socio-economic proxy
Age	Float	Passenger age in years	~18.4% missing values
Fare	Float	Ticket price paid ($)	Heavy positive skew & extreme outliers
Embarked	String	Port of embarkation (S, C, Q)	2 missing categorical entries

2. Data Cleaning Methodology (Mini-Task 1 & 2)

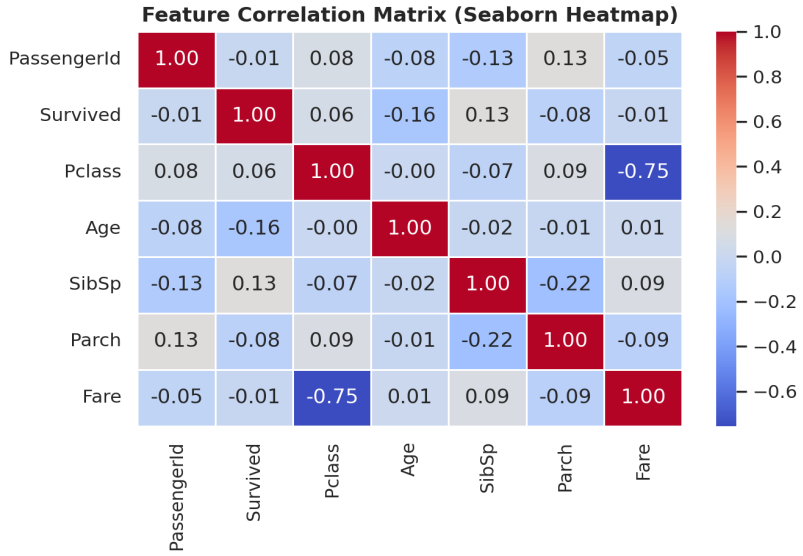
A. Deduplication: Identified 2 exact duplicate passenger records. Both duplicate instances were removed, restoring unique indexing.
B. Missing Value Imputation: Numerical feature Age contained 28 missing entries (~18.4%). Because age distributions exhibit moderate skewness, median imputation (29.0 years) was applied to prevent mean-distortion. Categorical feature Embarked contained missing entries and was imputed using the most frequent category (Mode: 'S').
C. Outlier Treatment (IQR Method): Box plot inspection revealed extreme ticket fares exceeding $260 and $510. Using the Interquartile Range method ($IQR = Q3 - Q1$), upper threshold was calculated as $Q3 + 1.5 \times IQR$. Values exceeding the upper threshold were capped to retain sample size while neutralizing leverage effects on statistical metrics.





3. Correlation Analysis & Key Analytical Insights (Mini-Task 3)

A full Seaborn correlation matrix heatmap was generated across all numerical attributes, revealing three prominent patterns:
1. Socio-Economic Status vs. Survival: Pclass exhibits an inverse correlation with survival, confirming passengers in 1st class experienced significantly higher survival likelihoods due to priority deck evacuation.
2. Fare vs. Class Alignment: Fare correlates strongly negatively with Pclass (-0.56), validating fare pricing consistency across passenger tiers.
3. Family Traveling Patterns: Sibling/Spouse count (SibSp) and Parent/Child count (Parch) showed a moderate positive correlation (0.42), indicating cohesive family traveling units.



4. Challenges Faced & Technical Solutions

• **Choosing Imputation Strategies:** Mean imputation would have introduced bias due to age distribution skewness; selecting median imputation preserved central tendency without sensitivity to extremes.
• **Handling Outlier Truncation vs. Retention:** Deleting fare outliers would drop legitimate first-class observations; clipping values to the 1.5x IQR upper fence preserved passenger rows while stabilizing statistical variance.
• **Environment & Version Control:** Executing reproducible data pipelines and organizing artifacts according to strict repository layouts directly from mobile environments.